

SIALKOT, Pakistan

WMO: 415990

Lat: 32.500N

Lon: 74.533E

Elev: 255

StdP: 98.30

Time Zone: 5.00 (E05)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.9	(c) 5.0	(d) 1.1	(e) 4.2	(f) 8.8	(g) 2.7	(h) 4.7	(i) 9.9	(j) 5.9	(k) 14.2	(l) 5.1	(m) 14.0	(n) 0.4	(o) 310	(p) 0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 12.6	(c) 42.2	(d) 22.5	(e) 40.6	(f) 22.7	(g) 39.0	(h) 22.6	(i) 28.9	(j) 33.8	(k) 28.4	(l) 33.3	(m) 28.0	(n) 32.8	(o) 2.6	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.8	24.6	31.7	27.2	23.7	31.6	26.8	23.1	31.4	96.6	33.6	94.4	33.3	92.3	32.5	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 5.5	(c) 4.6	(e) 1.6	(f) 44.9	(g) 0.5	(h) 1.2	(i) 1.2	(j) 45.7	(k) 0.9	(l) 46.4	(m) 0.6	(n) 47.1	(o) 0.2	(p) 47.9		
(a) 1.3	(b) 30.1	(c) 1.1	(e) 0.5	(f) 30.5	(g) 0.5	(h) 30.5	(i) -0.1	(j) 30.7	(k) -0.7	(l) 31.0	(m) -1.4	(n) 31.4	(o) -1.4	(p) 31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	11.5	15.1	19.9	26.1	30.7	32.2	30.4	29.6	28.5	24.6	18.4	13.3
	DBStd	7.51	2.31	2.48	3.21	3.14	2.56	2.78	2.26	1.84	1.80	2.55	2.15	2.45
	HDD10.0	14	11	1	0	0	0	0	0	0	0	0	0	3
	HDD18.3	509	211	94	21	0	0	0	0	0	0	0	26	158
	CDD10.0	4906	58	143	307	484	642	666	633	608	554	453	253	104
	CDD18.3	2359	0	4	69	234	384	416	374	350	304	194	28	1
	CDH23.3	31502	1	37	680	2877	5737	6477	5307	4686	3621	1826	242	11
	CDH26.7	17509	0	1	223	1608	3668	4233	2958	2373	1684	736	26	1
Wind	WSAvg	1.5	1.2	1.5	1.8	2.1	2.2	2.2	1.7	1.4	1.3	1.1	0.9	1.0
Precipitation	PrecAvg	970	37	39	44	24	19	76	275	255	121	13	9	21
	PrecMax	1887	115	136	148	72	45	163	694	918	480	48	44	113
	PrecMin	553	0	0	0	1	0	11	37	78	19	0	0	0
	PrecStd	404	36	34	41	19	14	46	178	214	106	14	13	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	26.1	34.1	40.2	44.1	44.2	40.8	36.2	35.8	34.1	28.8	24.8
		MCWB	14.7	17.2	20.4	21.9	21.3	22.7	25.9	28.0	26.8	25.3	19.6	16.0
	2%	DB	20.2	24.4	31.1	38.2	42.1	42.8	38.1	35.2	34.7	33.0	27.4	23.0
		MCWB	13.7	16.4	20.3	21.0	21.5	22.8	26.6	27.8	26.6	24.1	19.3	15.5
	5%	DB	18.9	22.9	29.0	36.8	40.2	41.1	36.4	34.8	33.9	32.0	26.1	21.2
		MCWB	12.8	15.6	19.0	20.5	21.4	23.1	27.1	27.7	26.6	23.5	18.5	14.6
	10%	DB	17.2	21.2	27.1	34.3	38.8	39.7	35.1	33.9	33.0	30.8	24.8	19.8
		MCWB	12.2	14.8	18.3	20.1	21.5	23.4	27.0	27.6	26.0	22.5	17.5	13.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	18.5	23.3	24.4	26.1	28.5	29.5	28.8	27.1	21.8	17.8	
		MCDB	20.3	24.2	30.1	36.2	37.2	35.4	34.8	33.8	33.7	31.9	26.8	22.3
	2%	WB	14.6	17.3	21.3	22.9	24.5	27.7	28.9	28.9	28.0	26.0	20.4	16.2
		MCDB	18.5	23.0	29.0	33.1	36.2	34.3	34.1	33.4	32.6	31.0	25.7	21.3
	5%	WB	13.5	16.4	20.0	21.8	23.5	26.9	28.4	28.5	27.3	25.1	19.3	15.2
		MCDB	17.5	21.5	27.0	33.0	35.6	34.2	33.5	33.0	32.1	29.8	25.0	19.9
	10%	WB	12.6	15.5	18.9	21.0	22.6	26.1	27.9	28.1	26.7	24.1	18.2	14.2
		MCDB	16.4	20.3	25.7	31.5	34.5	33.8	32.9	32.6	31.5	29.2	23.5	19.1
Mean Daily Temperature Range	5% DB	MDBR	11.0	11.5	12.6	14.6	14.7	12.6	8.3	7.4	8.7	12.0	13.1	12.6
		MCDBR	13.7	13.6	15.1	17.3	17.0	15.6	10.2	8.6	9.5	11.9	13.8	15.0
		MCWBR	7.9	7.0	6.5	5.9	5.6	5.2	4.1	3.9	4.2	5.0	7.1	8.6
	5% WB	MCDBR	12.1	12.2	13.9	15.3	14.9	11.1	8.6	7.8	8.6	10.1	12.9	14.1
		MCWBR	7.5	6.9	6.8	6.2	6.0	4.8	4.1	4.0	4.2	4.7	7.5	8.7
Clear-Sky Solar Irradiance	taub	0.461	0.490	0.499	0.542	0.596	0.686	0.734	0.665	0.574	0.571	0.529	0.481	
	taud	1.921	1.901	1.910	1.855	1.756	1.623	1.584	1.717	1.862	1.759	1.785	1.884	
	Ebn at Noon	717	745	780	766	727	662	627	664	707	660	639	666	
	Edh at Noon	155	175	186	205	228	260	269	232	193	199	177	154	
All-Sky Solar Radiation	RadAvg	2.96	3.85	5.21	6.29	7.15	6.96	5.76	5.47	5.45	4.93	3.76	3.08	
	RadStd	0.32	0.47	0.42	0.31	0.24	0.39	0.48	0.41	0.26	0.24	0.17	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (7 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page